Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The domain of $f(x) = \sqrt{\log \frac{1}{|\sin x|}}$ is

(A) R

(B) $R - \{-\pi, \pi\}$

(C) $R - \{x : x = n\pi, n \in Z\}$

(D) none of these

Q.2

The range of $f(x) = \frac{1}{1+\sqrt{x}}$ is equal to

(A) $\left(0, \frac{1}{2}\right]$

(B) $[0, 1]$

(C) $(0, 1]$

(D) none of these

Q.3

If $f(x) = \cos[\pi]x + \cos[\pi x]$, where $[y]$ is the greatest integer function of y, then $f\left(\frac{\pi}{2}\right)$ is equal to

(A) $\cos 3$

(B) 0

(C) $\cos 4$

(D) none of these

Q.4

The fundamental period of the function $f(x) = e^{\{4\{x\}+3\}}$, where $\{\cdot\}$ denotes the fractional part function is

(A) $\frac{1}{12}$

(B) 1

(C) $\frac{1}{4}$

(D) 4

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Q.5

If $f(x+y, x-y) = xy$ for all $x, y \in R$, then the value of $f(5,1)$ is

(A) 2

(B) 3

(C) 4

(D) 6

Q.6

The domain of function $f(x) = \log_e(x - [x])$ is (where $[x]$ is greatest integer $\leq x$)

(A) R

(B) $R - I$

(C) $(0, +\infty)$

(D) I

Q.7

Which of the following functions is inverse of itself in its domain?

(A) $f(x) = \frac{1-x}{1+x}$

(B) $f(x) = 5^{\log x}$

(C) $f(x) = 2^{x(x-1)}$

(D) none of these

Q.8

The fundamental period of the function $f(x) = |\sin x| - |\cos x|$, is equal to

(A) $\frac{\pi}{2}$

(B) π

(C) 2π

(D) none of these

Q.9

The exhaustive set of real values of x for which, the function, $f(x) = \sqrt{(x+2)(5-x)} - \frac{1}{\sqrt{x^2-4}}$, is a real valued function, is

(A) $[1, 2] \cup (2, 5]$

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(B) $(2, 5]$

(C) $[3, 4]$

(D) none of these

Q.10

The function $f(x) = \log\left(\frac{1+x}{1-x}\right)$ satisfies the equation

(A) $f(x+2) - 2f(x-1) + f(x) = 0$

(B) $f(x) + f(x+1) = f(x)f(x+1)$

(C) $f(x_1)\, f(x_2) = f(x_1 + x_2)$

(D) $f(x_1) + f(x_2) = f\left(\frac{x_1+x_2}{1+x_1x_2}\right)$

Q.11

The range of the function $f(x) = \sin[x], -\frac{\pi}{4} < x < \frac{\pi}{4}$, where $[x]$ denotes the greatest integer $\leq x$ is

(A) $\{0\}$

(B) $\{0, -1\}$

(C) $\{0, \pm \sin 1\}$

(D) $\{0, -\sin 1\}$

Q.12

The function $f(x) = \sin\left(\log\left(x + \sqrt{x^2+1}\right)\right)$ is

(A) even function

(B) odd function

(C) neither even nor odd

(D) periodic function

Q.13

Let $f(x) = x^2$ and $g(x) = 2^x$, then the solution set of $fog(x) = \text{gof}(x)$ is

(A) R

(B) $\{0\}$

(C) $\{0, 2\}$

(D) none of these

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Q.14

Let $g(x) = 1 + x - [x]$, [.] is GIF and $f(x) = \text{sgn}(x)$. Then for all x, $f\{g(x)\}$ is equal to

(A) x

(B) 1

(C) $f(x)$

(D) $g(x)$

Q.15

If the function $f : R \to A$ given by $f(x) = \frac{x^2}{x^2+2}$ is a surjection, then A is equal to

(A) R

(B) $[0, 1]$

(C) $(0, 1]$

(D) $[0, 1)$

Q.16

Which of the following are odd function?
(i) $f(x) = \frac{x}{1+|x|}$
(ii) $f(x) = x^2 \log\left(\sqrt{x^6+1} + x^3\right)$
(iii) $f(x) = \frac{2^x-1}{2^x+1} g(x)$, where $g(x)$ is an even function defined for $x \in R$
(iv) $\phi(x) = (f(x) + f(-x))(g(x) - g(-x))$

(A) (i) and (ii) only

(B) (i) and (iii) only

(C) (i) and (iv) only

(D) all of them

Q.17

If $f : R \to R$ where $f(x) = \frac{x^2+3x+6}{x^2+x+1}$ then $f(x)$ is

(A) many one and into

(B) many one and onto

(C) one-one and into

(D) one-one and onto

Q.18

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If $f(x) = \begin{cases} x \text{ when } x \text{ is rational} \\ 1-x \text{ when } x \text{ is irrational then } fof(x) \text{ is given as} \end{cases}$

(A) 1

(B) x

(C) $1+x$

(D) none of these

Q.19

If $f(x) = \frac{1}{1-x}(x \neq 1)$, then $f(f(f(x)))$ is

(A) a constant function

(B) equal to $f(x)$

(C) an identity function

(D) none of these

Q.20

If f be a function defined as $f(x) = \left[|x| \left[\frac{1}{|x|}\right]\right]$ when $|x| \neq \frac{1}{n}$ and $f(x) = 0$ when $|x| = \frac{1}{n}, n \in N$, $[x]$ denotes the integral point of x, then when $|x| \neq \frac{1}{n}$, $f(x)$ is

(A) 0

(B) 1

(C) n

(D) $\frac{1}{n}$

Q.21

Least positive period of the function $f(x) = [6x+7] + \cos \pi x - 6x$ where $[.]$ denotes greatest integer function, is

Q.22

If $f(x) = \cos(\log x)$, then $f(x)f(y) - \frac{1}{2}\left[f\left(\frac{x}{y}\right) + f(xy)\right]$ has the value

Q.23

If $f : R \to R$ is a function satisfying the property $f(x+1) + f(x+3) = 2, \forall x \in R$, then the least positive period of $f(x)$ is

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Q.24

If $f(x) = [x]^2 - \left[x^2\right]$, (where $[.]$ denotes the greatest integer function), and $x \in [0, 2]$ then the number of elements in the range of $f(x)$ is

Q.25

Let $f(x) = [x]$, the greatest integer less than or equal to x and $g(x) = \{x\}$, fractional part of x. The solution of the equation $4g(x) = i(x) + f(x); x > 0, i$ being identity function is $\mathrm{x} = \mathrm{p}/\mathrm{q}$ where p and q are coprime integers. Then the value of $\mathrm{p} + \mathrm{q} =$

Q.26

If $af(x) + bf\left(\frac{1}{x}\right) = x - 1, x \neq 0$ and $a \neq b$, then $[f(2)]$ for $(\mathrm{a}, \mathrm{b}) = (100, 50) =$____.
[.] is the GIF function.

Q.27

Let $f : R \to [0, \pi/2)$ (where R is the set of real numbers) be a function defined by $f(x) = \tan^{-1}\left(x^2 + x + a\right)$. If f is onto then 96 a equals

Q.28

The value of $|\alpha|$ for which the function $f(x) = 1 + \alpha x, \alpha \neq 0$ is the inverse of itself is (mark absolute value of answer)

Q.29

For a real number x, $[x]$ denotes the integral part of x. The value of $\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{100}\right] + \left[\frac{1}{2} + \frac{2}{100}\right] + \ldots\ldots + \left[\frac{1}{2} + \frac{99}{100}\right]$ is

Q.30

Let A be a set containing 5 distinct elements, then the total number of distinct functions from A to itself is

Answers & Solutions

Q.1 Answer:

$R - \{x : x = n\pi, n \in Z\}$

Solution:

$\log \frac{1}{|\sin x|} \geq 0$ **and** $|\sin x| \neq 0$
$\frac{1}{|\sin x|} \geq 1$ **and** $x \neq n\pi$**, where** $n \in Z$

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$|\sin x| \leq 1$ **and** $x \neq n\pi$, **where** $n \in Z$

$\Rightarrow$ **Domain** $= R - \{x : x = n\pi, n \in Z\}$

Q.2 Answer:

$(0, 1]$

Solution:

$0 \leq \sqrt{x} < \infty \Rightarrow 1 \leq 1 + \sqrt{x} < \infty \Rightarrow 0 < \frac{1}{1+\sqrt{x}} \leq 1 \Rightarrow 0 < f(x) \leq 1$

Q.3 Answer:

$\cos 4$

Solution:

$f\left(\frac{\pi}{2}\right) = \cos[\pi] \cdot \frac{\pi}{2} + \cos\left[\pi \cdot \frac{\pi}{2}\right] = \cos\frac{3\pi}{2} + \cos 4 = \cos 4$

Q.4 Answer:

$\frac{1}{4}$

Solution:

$f(x) = e^{\{4\{x\}+3\}} = e^{\{4(x)\}} = e^{\{4x\}}$

because $\{4\{x\} + 3\} = \{4\{x\}\}$ **and** $4\{x\} = 4x - 4\lfloor x \rfloor \Longrightarrow \{4\{x\}\} = \{4x\}$

Clearly $f\left(\frac{1}{4} + x\right) = f(x)$

Q.5 Answer:

6

Solution:

Let $x + y = 5$ **and** $x - y = 1$. **Then** $x = 3$ **and** $y = 2$

$\therefore \quad f(5, 1) = f(x + y, x - y) = xy = 3(2) = 6$

Q.6 Answer:

$R - I$

Solution:

$f(x) = \log(x - [x])$ **is defined if** $x - [x] > 0$

$\Rightarrow \quad \{x\} > 0 \quad \Rightarrow \quad x \notin I$

$\therefore \quad$ **Domain** $= R - I$

Q.7 Answer:

$f(x) = \frac{1-x}{1+x}$

Solution:

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Since fof $(x) = f(f(x)) = f\left(\frac{1-x}{1+x}\right) = \frac{1-\frac{1-x}{1+x}}{1+\frac{1-x}{1+x}} = x, \forall x$

$\therefore \quad fof = I \Rightarrow f$ **is the inverse of itself.**

Q.8 Answer:

π

Solution:

The period of $|\sin x|$ **is** π **and also period of** $|\cos x|$ **is** π**.**

So period of $|\sin x|_{-}|\cos x|$ **is** π**.**

Q.9 Answer:

$(2, 5]$

Solution:

The domain of $\sqrt{(x+2)(5-x)}$ **is the set of values of** x **for which** $(x+2)(5-x) \geq 0$

i.e., $x \in [-2, 5]$

The domain of $\frac{1}{\sqrt{x^2-4}}$ **is the set of values of** x **for which** $x^2 - 4 > 0$

i.e., $x \in (-\infty, -2) \cup (2, \infty)$**. The common values are** $(2, 5]$

Q.10 Answer:

$f(x_1) + f(x_2) = f\left(\frac{x_1+x_2}{1+x_1x_2}\right)$

Solution:

$f(x_1) + f(x_2) = \log\frac{1+x_1}{1-x_1} + \log\frac{1+x_2}{1-x_2} = f\left(\frac{x_1+x_2}{1+x_1x_2}\right)$

Because $\log\left(\frac{1+x_1+x_2+x_1x_2}{1-x_1-x_2+x_1x_2}\right) = \log\left(\frac{1+\frac{x_1+x_2}{1+x_1x_2}}{1-\frac{x_1+x_2}{1+x_1x_2}}\right)$

Q.11 Answer:

$\{0, -\sin 1\}$

Solution:

When $-\frac{\pi}{4} < x < 0 \quad \Rightarrow \quad -1 < -\frac{\pi}{4} < x \quad \Rightarrow \quad [x] = -1$

$\Rightarrow \quad y = \sin(-1) = -\sin 1$**, when** $0 \leq x < \frac{\pi}{4} < 1 \Rightarrow [x] = 0$

$y = \sin 0 = 0$

Q.12 Answer:

odd function

Solution:

$f(x) = \sin\left(\log_e\left(x + \sqrt{x^2+1}\right)\right)$

$\Rightarrow \quad f(-x) = \sin\left[\log_e\left(-x + \sqrt{1+x^2}\right)\right]$

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$\Rightarrow f(-x)=\sin\log_e\left(\left(\sqrt{1+x^2}-x\right)\left(\frac{\left(\sqrt{1+x^2}+x\right)}{\left(\sqrt{1+x^2}+x\right)}\right)\right.$

$\Rightarrow f(-x)=\sin\log_e\left[\frac{1}{\left(x+\sqrt{1+x^2}\right)}\right] \Rightarrow f(-x)=\sin\left[\log_e\left(x+\sqrt{1+x^2}\right)^{-1}\right]$

$\Rightarrow \quad f(-x)=\sin\left[-\log_e\left(x+\sqrt{1+x^2}\right)\right] \Rightarrow f(-x)=-\sin\left[\log_e\left(x+\sqrt{1+x^2}\right)\right.$

$\Rightarrow \quad f(-x)=-f(x)$

$\therefore f(x)$ **is odd function.**

Q.13 Answer:

$\{0,2\}$

Solution:

$\text{fog}(x)=f(g(x))=f\left(2^x\right)=\left(2^x\right)^2=2^{2x}$ **and** $\text{gof}(x)=g(f(x))=g\left(x^2\right)=2^{x^2}$

Thus the solution of $2^{x^2}=2^{2x}$ **is given by** $x^2=2x$ **which is** $x=0,2.$

Q.14 Answer:

1

Solution:

$\therefore \quad g(x)=1+x-[x]=1+\{x\}\geq 1 \quad \forall x\in R$

by definition of $f(x)=1, x>0$

$f(g(x)\geq 1)=1$

Q.15 Answer:

$[0,1)$

Solution:

Range of $\frac{x^2}{x^2+2}=A \quad \Rightarrow \quad x^2=\frac{2A}{1-A}\geq 0 \quad \Rightarrow 0\leq A<1$

Q.16 Answer:

all of them

Solution:

(i), (iii) and (iv) are clearly odd, for (ii) we use

$\left(\sqrt{x^6+1}+x^3\right)\left(\sqrt{x^6+1}-x^3\right)=1$

Q.17 Answer:

many one and into

Solution:

Clearly domain of $f(x)$ **is all real numbers**

$f'(x)=\frac{-2x^2+10x-3}{(x^2+x+1)^2}=\frac{2\left(x^2-10x+3\right)}{(x^2+x+1)^2}$

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Now since discriminant of $2x^2 - 10x + 3$ is positive thus $f'(x)$ will not remain its sign throughout the domain.

Hence $f(x)$ is many one.

Also $\lim_{x\to\pm\infty} f(x) = 1$ thus f(x) is into.

Q.18 Answer:

x

Solution:

$$\text{fof} = \begin{cases} f(x) \text{ when } f(x) \text{ is rational} \\ 1 - f(x) \text{ when } f(x) \text{ is irrational} \end{cases}$$

$$= \begin{cases} x \text{ when } f(x) \text{ is rational} \\ 1 - (1 - x) \text{ when } f(x) \text{ is irrational} \end{cases} = x$$

Q.19 Answer:

an identity function

Solution:

$f(x) = \frac{1}{1-x}, f(f(x)) = \frac{x-1}{x}$ **and** $f(f(f(x))) = x$**.**

Q.20 Answer:

0

Solution:

$|x| \neq \frac{1}{n} \Rightarrow \frac{1}{|x|} \neq n \Rightarrow \frac{1}{|x|} = n + F$ **where)** $< F < 1$

$\therefore |x| \left[\frac{1}{|x|}\right] = \frac{1}{n+F} \cdot n = \frac{n}{n+F} < 1 \quad \Rightarrow \quad f(x) = \left[|x| \left[\frac{1}{|x|}\right]\right] = 0$

Q.21 Answer:

—

Solution:

$f(x) = [6x + 7] + \cos \pi x - 6x = [6x] + 7 + \cos \pi x - 6x$

$f(x) = [6x] - 6x + \cos \pi x + 7$

$f(x) = -\{6x\} + \cos \pi x + 7$

Period of $\cos \pi x = 2$ **and period of** $\{(6x)\}$ **is** $\frac{1}{6}$

and LCM of 2 and $\frac{1}{6}$ **is 2.**

So period of $f(x)$ **is 2.**

Q.22 Answer:

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Solution:

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$f(x)f(y)-\frac{1}{2}\left[f\left(\frac{x}{y}\right)+f(xy)\right]=\cos(\log x)\cos(\log y)-$
$\frac{1}{2}\left[\cos\left(\log\left(\frac{x}{y}\right)\right)+\cos(\log(xy))\right]$
$=\left(\log(\log x)\cos(\log y)-\frac{1}{2}[\cos(\log x-\log y)+\cos(\log x+\log y)]\right.$
$=\left(\cos(\log x)\cos(\log y)-\frac{1}{2}[2\cos(\log x)\cos(\log y)]=0\right.$

Q.23 Answer:

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Solution:

$f(x+1)+f(x+3)=2, \quad f(x+3)+f(x+5)=2$
$f(x+1)-f(x+5)=0 \quad \Rightarrow \quad f(x+1)=f(x+5) \quad \Rightarrow \quad f(x)=f(x+4)$

Q.24 Answer:

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Solution:

$$f(x)=[x]^2-\left[x^2\right]=\begin{cases}0, & 0\le x<\sqrt{2}\\ -1, & \sqrt{2}\le x<\sqrt{3}\\ -2, & \sqrt{3}\le x<2\\ 0, & x=2\end{cases}$$

$\therefore$ **Range of** $f(x)$ **is** $\{-2,-1,0\}$.

Q.25 Answer:

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Solution:

As $\{x\}=x-[x]$**, the given equation can be written as**
$4(x-[x])=x+[x]\Rightarrow 3x=5[x]\Rightarrow x=\frac{5[x]}{3}$
Next using $[x]=x-\{x\}$**, we can write the given equation is**
$4\{x\}=x+x-\{x\}\Rightarrow x=\left(\frac{5}{2}\right)\{x\}$
As $0\le\{x\}<1$**, we get** $0\le x<\frac{5}{2}$
Thus $0\le\frac{5[x]}{3}<\frac{5}{2}\Rightarrow \quad 0\le[x]<\frac{3}{2}$
Since $[x]\in I$ **so** $[x]=0$ **or 1 thus** $3x=0$ **or** $3x=5\Rightarrow x=\frac{5}{3}$
But $x>0$ **so** $x=\frac{5}{3}$

Q.26 Answer:

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Solution:

Given, $af(x)+bf\left(\frac{1}{x}\right)=x-1,,x\neq 0,a\neq b \quad$ **....(i)**
$\Rightarrow af\left(\frac{1}{x}\right)+bf(x)=\frac{1}{x}-1$ **...(ii)**

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$a \cdot (\mathrm{i}) - b.(\mathrm{ii})$
$\Rightarrow \quad \left(a^2 - b^2\right) f(x) = a(x-1) - b\left(\frac{1}{x} - 1\right) \Rightarrow \left(a^2 - b^2\right) f(2) = a + \frac{b}{2} = \frac{2a+b}{2}$
$\Rightarrow \quad f(2) = \frac{2a+b}{2(a^2-b^2)}$

Q.27 Answer:

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Solution:

F **is onto** $\Rightarrow$ **Range of** f **is** $[0, \pi/2) \Rightarrow \quad 0 \leq \mathrm{x}^2 + x + \mathrm{a} < \infty, \forall x \in \boldsymbol{R}$
$\Rightarrow \quad 0 \leq \left(x + \frac{1}{2}\right)^2 + a - \frac{1}{4} < \infty, \forall x \in \boldsymbol{R} \Rightarrow \quad \mathrm{a} = 1/4$

Q.28 Answer:

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Solution:

$f^{-1}(x) = \frac{x-1}{\alpha}$
Now, $f(x) = f^{-1}(x) \Rightarrow \alpha = -1$

Q.29 Answer:

—

Solution:

$\therefore$ **denotes the integral part of** x**.**
Hence, after term $\left[\frac{1}{2} + \frac{50}{100}\right]$**, each term will be one.**
Hence the sum of given series will be 50.

Q.30 Answer:

—

Solution:

Each element in set A **has** m **options in it's images, so number of mappings** $= m \times m \times \ldots \times m = m^m$
m **times will be equal to**
$5 \times 5 \times 5 \times 5 \times 5 = 3125$

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